Lavanya Bhatnagar

Roll No.:230003035
B.Tech
Mechanical Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology Indore	8.5 (Current)	2023–Present
Senior Secondary	CBSE Board	96.4%	2023
Secondary	CBSE Board	95.4%	2021

Experience

- **IITI Drishti CPS Foundation** — *May 2025 – Jul 2025*
  Summer Internship for Data-Driven Healthcare Innovations Intern — IIT Indore
  - Collaborated with a team of 4 members to engineer a drone prototype for medical delivery to remote locations, optimising for the challenging conditions of remote village health centres.
  - Developed, simulated and integrated real-time telemetry within an autonomous drone architecture using a Pixhawk flight controller.
  - Designed and implemented a payload carrier with hardware-in-the-loop, validated robustness for remote healthcare delivery.

Projects

- **NIDAR – Coordinated Dual-Drone Perception System for Disaster Response** — *Sep 2025 – Dec 2025*
  Software & Autonomy Lead | National Drone Competition (MeitY) — GitHub
  - Developed a real-time aerial perception pipeline using YOLO-based object detection to reliably identify humans and relevant objects in outdoor, disaster-like environments.
  - Fused onboard camera feeds with GPS telemetry to enable vision-triggered geotagging of detected humans, supporting situational awareness and rescue-oriented decision-making.
  - Integrated and optimized the perception stack on NVIDIA Jetson Orin Nano, balancing inference latency, detection accuracy, and onboard power constraints.
  - Implemented a vision-to-action feedback loop, where perception outputs directly influenced drone behavior, logging, and mission context, demonstrating closed-loop autonomous operation.
- **Autonomous Mobile Rover – Warehouse Inventory Management** — *Nov 2025 – Dec 2025*
  Inter IIT Tech Meet 14.0 | Eternal Company PS | Bronze Medal (3rd out of 23 IITs)
  - Deployed an autonomous ground robot for warehouse inventory auditing, integrating SLAM-based mapping, AMCL localization, autonomous navigation, and vertical QR-based scanning up to 1.8 m.
  - Implemented a ROS 2 Nav2 navigation pipeline with LiDAR–encoder fusion, including custom costmap layers and inflation tuning, achieving ±10 cm shelf alignment accuracy in aisle-like environments.
  - Developed and integrated a hybrid obstacle avoidance system, fusing LiDAR-based global costmaps with ultrasonic-based local obstacle layers to handle low-height and dynamic obstacles in real time.
  - Built a web-based HMI using rosbridge/WebSockets, enabling live robot monitoring, manual overrides, and structured inventory data logging with robot pose metadata.
- **ATTIS – Acoustic Tracking and Target Identification System** — *Oct 2025 – Nov 2025*
  Instrumentation and Controls Project
  - Built an end-to-end acoustic perception pipeline using a three-microphone array for real-time sound source localization under embedded constraints.
  - Implemented ESP32-based multi-channel audio acquisition, streaming synchronized data over serial and applying DSP filtering and frequency thresholding to isolate salient sound events.
  - Devised direction-of-arrival estimation logic using inter-microphone signal differences and implemented a multi-threaded processing architecture for concurrent signal processing and data logging.
  - Validated system performance through controlled experiments, quantifying angular accuracy, range limitations, and noise sensitivity, and documenting key system trade-offs.
- **LifeBlink – Smart Eyewear for the Impaired** — *Jan 2025 – Feb 2025*
  Dr. Vibhor Pandhare, Asst. Professor, IIT Indore — GitHub
  - Designed non-invasive smart eyewear for communication and health monitoring in speech-impaired individuals.
  - Integrated PPG and EOG sensors for blood pressure and eye-blink/iris direction detection.
  - Developed LSTM-based classification and deployed model on ESP32 microcontroller with cloud integration.

Technical Skills

- **Programming:** Python, C/C++, MATLAB
- **Frameworks/Libraries:** ROS2, OpenCV, YOLOv8, Dronekit, TensorFlow*
- **Perception:** Computer Vision, Sound Signal Processing, Point Cloud Data Handling
- **Operating Systems:** Windows, Linux
- **Simulation and Modeling:** Gazebo, PX4-QGC, Simulink*
- **Hardware and Embedded Systems:** Arduino, Raspberry Pi, Teensy, Jetson Nano, ESP32, sensor integration and fusion
- **Data Science and ML:** Machine Learning, Data Visualization, EDA
- **Software and Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Docker*

** Elementary proficiency*

Key courses taken

Mathematics: Linear Algebra, Basic Calculus, Differential Equations, Numerical Methods

Introduction to Robotics: Body Frames and Orientation, Kinematics, Dynamics, Sensors, Actuators

Control Systems: Flight Mechanics and Control Systems, Drone Systems and Controls (Prof. Suresh Sundaram, IISc)

Data Analytics: Introduction to Data Analytics, Python and Machine Learning

Simulation and Modeling Tools: MATLAB Onramp, Simulink Onramp

Positions of Responsibility

- **Head of Autonomy**, Robotics Club, IIT Indore *Apr 2025 - present*
- **Head of Classical Dance Group**, Dance Club, IIT Indore *Apr 2025 - present*
- **Head of Hospitality**, E-Summit, IIT Indore *Jul 2024 - Oct 2024*
- **Core Member**, Debating Society, IIT Indore *Aug 2024 - May 2025*
- **USG Executive Board Member**, Model United Nations 8.0, IIT Indore *May 2024 - Jan 2025*

Achievements & Honors

- **Bronze Medal**, Inter IIT Tech Meet 14.0 – Autonomous Mobile Rover with Vertical Scanning & Warehouse Inventory Management – Eternal Company Problem Statement | 3rd out of 23 IIT teams *2025*
- **Qualified Pre-Finals**, NXP Artificial Intelligence in Mobility (AIM), Noida *2024*

Conferences & Technical Exposure

- **AIR 2025 – Advances in Robotics Conference** *Jul 2025*
  Attended technical talks, research presentations, and industry sessions on autonomous systems, perception, and robotic deployment